

Course Project One Pager

Team Details:

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Project Title:

Explainable Root Cause Analysis for Graph Neural Network Traffic Prediction Failures via Topology-Aware Attribution

Possible Dataset and Source:

Primary: Abilene Network Dataset (12 nodes, 24 weeks of real traffic data)

Source: SNDlib - <http://sndlib.zib.de/>

Secondary: GÉANT Network Dataset (23 nodes, European backbone traffic)

Source: SNDlib - <http://sndlib.zib.de/>

Uniqueness of Topic:

- No existing work combines explainability, root cause analysis, and topology-aware attribution specifically for network traffic prediction failures.
- While GNN explainability exists for correct predictions, diagnosing why predictions fail and identifying responsible network components is unexplored in computer networks.
- Current traffic prediction models fail silently during network disruptions without providing actionable insights to operators about which nodes/links caused the failure.
- Existing root cause analysis (RCA) work focuses on microservices and cloud systems, not computer network traffic prediction.

Our Learning Goal:

The main learning objective is to devise an explainable AI framework that leverages Graph Neural Networks and topology-aware attribution methods for diagnosing prediction failures in network traffic forecasting. We bridge the gap between model explainability and operational network management, where the operator understands why the prediction is failing, which network component is failing the prediction, and how to intervene proactively. This project contributes to a deeper understanding of GNN interpretability, root cause analysis techniques, and their practical application in critical infrastructure systems.

Why We Think This Project is Important:

- Operational criticality: Network operators currently deploy GNN-based traffic predictors without understanding when or why they fail, leading to potential service disruptions.
- Trust and deployment: Explainability is essential for deploying AI in critical infrastructure. Network engineers need to understand model failures to trust and act on AI recommendations.
- Proactive management: Identifying root causes enables preventive maintenance - operators can address problematic nodes/links before failures cascade through the network.
- Real-world impact: Computer networks underpin internet infrastructure, cloud services, and enterprise operations. Reliable traffic prediction with failure diagnosis directly impacts service quality for billions of users.

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Reference Papers:

	Paper Title	Link to Paper
1	GNNExplainer: Generating Explanations for Graph Neural Networks	https://arxiv.org/abs/1903.03894
2	TraffExplainer: A Framework toward GNN-Based Interpretable Traffic Prediction	https://ieeexplore.ieee.org/document/10680338
3	Graph WaveNet for Deep Spatial-Temporal Graph Modeling	https://arxiv.org/abs/1906.00121
4	Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting	https://arxiv.org/abs/1707.01926